European Journal of Purchasing and Supply Management 1994 **1**(1) 27–35

The changing role of purchasing: reconsidering three strategic issues

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The competitive pressures that act upon firms have led to major changes in the way in which purchasing is perceived – from a money-saving activity to a resource-planning function. In this paper the authors discuss some of the changes that purchasing has undergone and identify key concerns for the future, as purchasing moves towards a new strategic role.

Keywords: purchasing strategy, buyer–seller relationships, supply management

During the last 20 years a new view of purchasing has gradually emerged. From being considered a clerical function – with the ultimate purpose of buying as cheaply as possible – it is today regarded in many companies as a major strategic function. This new attitude towards purchasing is not surprising, as purchasing is responsible for more than half the total costs in many companies. Nevertheless, the scale of the recent change is so considerable that an analysis of its development is of interest. This paper draws on our own research and that of many others. Our focus is manufacturing and assembly industry.

Let us begin with a quotation that gives a representative statement on the traditional view of efficiency in purchasing:

> Price has been the principal yardstick by which manufacturers have traditionally selected their suppliers. By spreading their purchases among several suppliers, it is argued, manufacturers can achieve the cheapest price and the greatest assurance of a secure flow of material. (Dillforce, 1986, p 3)

Such purchasing behaviour does not allow for making direct use of the total resources of the suppliers. This can be a considerable drawback, according to the analysis provided by, for example, Axelsson and Håkansson (1984) and Spekman (1985). During the last decade, however, many companies have changed their behaviour in the directions proposed (see for example Carlisle and Parker, 1989; Gadde and Håkansson, 1993; Lamming, 1993).

In 1987, Morgan observed a tendency among customers to move from an arm's-length relationship (a number of competing suppliers) towards single sourcing and even to 'alliances'. The last type of relationship involves a deepened cooperation between the customer and a specific supplier. Similar frameworks have been presented by Frazier *et al* (1988) and Lamming (1993). Frazier *et al* have developed a conceptual model where the traditional view is called *market exchange*. The deeper relationships are then characterized as *relational exchange* and *just-in-time exchange*. Lamming's discussion is based on empirical observations from the automobile industry. This model takes a dynamic view and identifies a four-phase development process from the traditional model (prior to 1975) to the partnership model (after 1990).

The strategic purchasing context for the partnership model is very different from that of the traditional model, and similarly for just-in-time exchange versus market exchange. The aim of our contribution, therefore, is to analyse the changing role of purchasing in terms of purchasing strategy and

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purchasing behaviour. We shall discuss the impact of the new view in three dimensions of purchasing strategy.

The major strategic decision is always whether to buy or to rely on in-house production. Consequently, the first strategic issue is related to the question of *make or buy*. If the role of purchasing is changing, the nature of the make-or-buy decision might be affected.

The second strategic choice is to decide on the *supply-base structure*. This issue regards the number of suppliers to use, as well as how the suppliers should relate to one another. The third strategic aspect deals with the nature of the *customer–supplier relationship*. When buyer–seller relationships turn into alliances, partnerships or just-in-time exchanges, the suppliers become more or less parts of the customer's production system. We discuss the benefits that can be obtained from this kind of close relationship.

Make or buy

The first strategic issue deals with the decision of whether to make or to buy. This question has been a major topic ever since industrial activities were established but it does not seem to have been considered as a strategic problem by manufacturing firms. Culliton (1942) concluded that most managers, when they were asked about their make-or-buy problem, said they had none. According to Culliton, however, that answer should be considered as representing a lack of insight into strategic issues, rather than the absence of the problem. The relevance of this aspect seems to have been neglected even in more recent times: see for example Janch and Wilson (1979), Leenders and Nollet (1984), and Ford *et al* (1993). These authors conclude that make-or-buy problems have generally been ignored by top management. Where they have been considered, they seem to have been handled by purchasing departments and based on short-term cost criteria, rather than on strategic analysis.

In spite of the perceived absence of strategic decisions, it is possible to identify a clear trend over time towards the increasing importance of 'buy'. In 1942 Culliton argued that, in general, it seemed that 'buying is preferable to making'. The reasons for his conclusion, however, appear to be the disadvantages associated with a 'make' strategy, owing to rapid changes in the market, and the lack of flexibility that characterizes in-house production. Hayes and Abernathy (1980) agree with Culliton in advocating a 'buy' strategy. Their main reason is the risks associated with being locked into a specific technology, which might be challenged by new developments. A company with asset-specific investments, they argue, may have problems in maintaining its innovativeness. The authors therefore warn companies against too high a degree of vertical integration. Dirrheimer and Hübner (1983) analysed the vertical integration in the automobile industry and discovered a considerable variation between different companies and different countries. For all manufacturers analysed, however, vertical integration had decreased during the previous five-year period.

In total, therefore, as the levels of vertical integration have increased, suppliers appear to have become more and more important as sources of resources. This is illustrated by examples in Gadde and Håkansson (1993). In a study of four large multinationals in the Swedish engineering industry, it was shown that components and systems purchased from outside suppliers accounted for about two-thirds of the turnover. In the construction industry, it was found, the proportion is even higher. The situation also seems to be similar for small and medium-sized firms. In a study of 123 Swedish companies, purchasing accounted for more than 40% of the turnover for 70% of the companies. For almost every fifth firm, the proportion extended to 70% (Håkansson, 1989).

The figures also seem to be representative for a general trend in other countries (Ford *et al*, 1993) and other industries (Kumpe and Bolwijn, 1988). These findings show that many manufacturing companies are increasingly relying on competent suppliers, which have been able to contribute to the customer's efficiency in production as well as in R&D.

It is possible, therefore, to identify a decreasing rate of vertical integration. A strategic shift of this kind can lead both to advantages and to disadvantages. One interesting question is whether this trend will continue. Miles and Snow (1986) predict that it will. In their opinion, the changing environmental conditions and new competitive forms will result in industries' being characterized by completely new organizational forms in the future. They suggest that the functions necessary in an industrial system (such as design, production and marketing) will be performed by specialized companies, each one undertaking separate functions. Such dynamic networks are supposed to be fairly loose coalitions intertwined by 'brokers' responsible for the integration of the functions. In this integration, exchange of information will be a crucial determinant of efficiency and effectiveness, as the network governance is based on market mechanisms. Such an industrial system is characterized by a lack of vertical integration.

Kumpe and Bolwijn (1988), in contrast, argue that the degree of vertical integration will increase in the future, because value chains today are characterized by an imbalance in the distribution of profits. Enterprises in the final stages of the production chain (assembly and marketing/distribution) usually tend to be profitable, while profits in the earlier stages (such as component production) are generally

substantially less profit-making. For the value chain as a whole to be competitive, however, there is a great need for investments in these stages as well. There is an obvious risk, then, that the investments necessary for long-term competitiveness (in design, product development and component production) will never be undertaken, because suppliers might be unable to raise the necessary financial resources. Therefore, Kumpe and Bolwijn argue, firms in the final stages of the value chain will be forced to integrate backwards and provide these resources; otherwise suppliers might be driven out of business, implying severe problems for the customer. A strategy aiming at an even lower degree of vertical integration, which might be efficient in the short term, is therefore considered to be a disaster in the long run.

We have thus been able to identify two contradictory views of the future of vertical integration. The divergent opinions can partly be explained by their focus on the formal degree of integration: that is, on ownership ties. As we have seen, vertical integration in this respect has decreased over time. It has been replaced, however, by informal arrangements that keep the industrial networks together. This is what is usually called 'quasi-integration' (Blois, 1972). Quasi-integration can take various forms and include, among other things, customer investments in production tools, joint product development and various forms of financial support. It is apparent that these informal mechanisms are increasing in importance over time. Therefore, the divergent opinions held by Kumpe and Bolwijn and Miles and Snow need not be as conflicting as they appear at first sight. Through quasi-integration arrangements customers can provide suppliers with support, without moving back to ownership relations. However, buying firms will not be free to change suppliers as easily as indicated by Miles and Snow. Investments undertaken on either side of the dyad will restrict the opportunities to change supplier.

We do not foresee any major strategic shift in the view of make-or-buy decisions. Our opinion is that the observed tendency – towards buying more from outside suppliers – will continue. The driving forces towards increasing specialization will still be strong in the future, and will favour outsourcing not only of components but also, at an increasing rate, of systems of components, design and product development. One important implication following from our discussion, however, is that make-or-buy decisions should be given more strategic attention than in the past. When most of the resources of a company are provided by outside suppliers it is necessary for these decisions to be given top priority. This need is heightened because make-or-buy decisions are increasingly characterized by technical complexity. Venkatesan (1992) has found that the aim of preserving jobs has been one important determinant of make-or-buy decisions. The effect, in many cases, has been insourcing of parts that are easy to manufacture and outsourcing of components and systems that are more complicated. In the long run, such a strategy might erode the capability of the firm. According to Venkatesan, therefore, the management challenge is to identify the really strategic components among the thousands of parts that they know mostly in terms of cost, rather than function or importance to the product.

Design of the supply-base structure

The issues regarding the supply-base structure can be divided into two strategic aspects. One has to do with the *number of suppliers*, the other with *the way suppliers are organized*. The number of suppliers has always been an important aspect of purchasing strategy. As mentioned above, the traditional view of purchasing meant that there was a group of suppliers eagerly competing with one another – mainly in terms of price: thus the more suppliers a company had in its supply base the better. We know that this view of efficiency has been increasingly questioned and we discuss the implications below. One of the reasons for the changing view is the insight into the advantages that can be obtained from more cooperative relationships with a reduced supply base. Activities that deepen individual relationships will also have an effect on the supply-base structure as a whole. In fact, many companies today have clear strategic aims for the shape of the supply-base structure. This is a growing dimension of purchasing strategy.

Number of suppliers

The choice between single sourcing and multiple sourcing is a classical issue in purchasing strategy. The established criterion of efficiency has often resulted in multiple sourcing, as supplier competition has been given priority. By promoting competition among suppliers, customers are expected to be given better control of price levels as well as more reliable supply through diversification of risks.

It is obvious that purchasing strategies have undergone considerable changes in this respect during the 1980s. Newman (1988) has identified a clear trend towards single sourcing. The significance of the changes differs between industries. It seems to be most obvious in the automotive industry. The North American motor manufacturer Chrysler decreased its number of suppliers of wiring harnesses from fourteen to three and its paint suppliers from five to two: one for the US plants and one for the Canadian plants (Raia, 1988). General Motors reduced its supplier base for its Quad-four engine from 140 to 69 through single sourcing (Offodile and Arrington, 1992). A US survey of more than 1000 suppliers showed that the average

number of firms competing with the average supplier to produce the same product for a given customer decreased from 2.0 to 1.5 in five years (Helper, 1991).

Such changes for specific components and firms have substantial effects on the supplier structure as a whole. Considerable changes in the total number of suppliers have been reported for Ford US, from 3200 in 1981 to 2100 in 1987 (Burt, 1989), and Chrysler, from 3000 suppliers on the roll only a few years ago to just over 1000 today (Raia, 1993). Similar changes are evident in other industries and companies: in 1981 Rank Xerox had almost 5000 suppliers, but six years later the number had been reduced to just over 300 (Morgan, 1987).

These observations contrast sharply with the traditional view of purchasing efficiency. Newman (1988) states that only a decade ago a purchasing strategy relying on single sourcing would have been characterized as an 'invitation to disaster'. At that time customers implementing single sourcing were expected to lose opportunities for price control, as well as diversification of risks. The reason for looking at this trade-off through other glasses is the changing role of purchasing. Today it is possible, in fact, to argue that single sourcing leads to an increase in the reliability of supply. Purchasing firms that reduce the number of suppliers and try to strengthen their relationships with the remaining ones might be able to establish very efficient logistical systems together with their suppliers. We discuss this below when dealing with the quality of supplier relations. Regarding price control, Newman (1988) argues that the price competition perceived when dealing with multiple sourcing can often be illusory. Price is only one of the costs affected and changed, when purchasing activities are handled differently. A number of indirect costs are also affected, as discussed below.

Organization of the supply-base structure

The second dimension of the supply-base structure has to do with organization of suppliers. The effects of various organizational forms are illustrated in a study of supplier relationships in the automobile industry (Gadde and Grant, 1984). The number of suppliers dealing directly with automobile manufacturers showed substantial variability. General Motors, in their US operations, used around 3500 suppliers, while Volvo Car Corporation relied on 800 suppliers. The difference between GM and Volvo can be explained by scale factors such as number of plants and number of cars produced. The corresponding figures for manufacturers from Japan, however, did not fit into this pattern. Only around 200 suppliers delivered directly to Toyota and Nissan in spite of the fact that the production output of these companies was closer to that of GM than that of Volvo.

The main reason for these deviations was shown to be the different ways the companies organized the supply-base structure. GM and Volvo, in fact, had no organization at all. The strategy of supporting competition among suppliers through multiple sourcing had resulted in fairly 'wild' structures. Toyota and Nissan had organized their suppliers in hierarchies. Only the suppliers on the first level delivered directly to the customer. These first-tier suppliers were made responsible for just-in-time deliveries. They had also become more 'system' than 'component' suppliers over time, and were to an increasing extent responsible for product development. Furthermore, they were responsible for the activities of the suppliers on the other levels in the structure. Nishiguchi (1987, 1993) provides a detailed discussion of these systems and their characteristics.

The first analysis of the Japanese supplier structures concluded that customers were relying on single sourcing. More recent research, however, has shown that this is true for large complex systems that require massive investments in tools. In other supply situations, however, two or more vendors can be used (Womack *et al*, 1990). The same observation is further elaborated by Richardson (1993) who characterized the purchasing strategy used by Japanese auto-manufacturers as *parallel sourcing*. This system is observed when the car manufacturer produces a number of models at different plants using a sole source for a component of one model at one assembly plant, while another source is used at another plant. A second aspect of parallel sourcing is that some components can be common to various models and may have multiple sources, while they may still be sole sourced for a particular model.

When the number of suppliers is reduced, the customer becomes more dependent on individual suppliers. The traditional view of efficiency recommends that customers avoid such dependences. In spite of this we have seen firms turning in this direction. The underlying reason is that reducing the number of counterparts is a prerequisite for improved and deepened supplier relationships.

The nature of customer–supplier relationships

The main reason for reducing the number of suppliers has been to provide opportunities for deeper cooperation with selected individual firms. Two driving forces can be identified in this process. One is reduction of costs: the potential for rationalization through a close relationship. The other is the possible exploitation of supplier resources in order to improve technical development: that is, development through a close relationship. We shall now discuss these two dimensions of increasing quality in supplier relations.

Cost rationalization effects can be obtained in several ways. A number of indirect costs can be

affected through a deeper cooperation with suppliers. We shall discuss these effects in terms of administrative costs, production costs and costs related to the material flow. Costs of R&D and product development can also be affected. These are discussed in the section on development cooperation.

Rationalization of administrative costs

The flow of administrative information between customer and supplier provides substantial potential for rationalization. In Gadde and Håkansson (1993) a number of examples are presented. One of them deals with a customer whose ambition was to decrease the paper flow in one of the supplier relationships. By eliminating purchase orders and introducing daily deliveries with monthly invoicing, it was possible to decrease the number of documents per transaction from seventeen to three. Another illustration of the substantial administrative costs is a big Swedish company in the construction industry that receives around 1.2 million invoices a year. Internal calculations indicate that the cost for handling one invoice is more than 300 SEK (about US$40). Clearly, the administrative costs represent an enormous potential for rationalization.

A powerful tool for attacking these costs has been obtained through the development of information technology. Enquiries, orders and invoices can be transmitted today, very quickly and accurately. In addition to reducing administrative costs, such arrangements provide purchasing officers with more time for working with strategic questions. An analysis of the impact of information technology on purchasing is presented in Dubois *et al* (1989).

Rationalization of production costs

One way of affecting production costs is to combine in-house production capacity and capability with the corresponding supplier resources. This is the general make-or-buy problem as already discussed. By moving production activities from one firm to another it is possible to increase efficiency. It is clear, however, that such changes in the division of labour can be handled in a more sophisticated way within more intimate customer–supplier relationships. In long-term relationships successive adaptations will enhance performance. Another effectiveness measure may be coordination amongst suppliers: for example, by increasing the cooperation between them. Marler (1989) exemplifies this by illustrating how Ford told all their suppliers of door components that they were no longer interested in purchasing individual components. The suppliers were encouraged to do what they could to form alliances and then tender for complete door systems.

The Ford example illustrates the increasing interest of many automotive firms in purchasing complete systems rather than separate components. One implication of this general trend is a development towards networks of highly specialized production units. As shown by Dubois and Håkansson (1993) the activity structures of such networks can be combined in a number of different ways. This is another reason why the make-or-buy decisions must be given increasing strategic attention.

Rationalization of material flow costs

Costs related to material flow include the costs of handling goods, costs of keeping inventories, and costs of capital. A reduction in these costs is probably the most significant advantage that can be obtained through closer supplier relationships. Many firms have achieved substantial improvements in efficiency by reducing stocks of input goods and work in process. Activities of this kind were originally inspired by observations from Japan:

> It only takes 10 minutes inside an assembly plant in Japan to realize that relationships with suppliers are very different. The visitor accustomed to the loading docks, the large storage areas and the large incoming inspection area, typical of US plants, is likely taken aback by the stocking of Japanese assembly lines. Trucks from suppliers back up through large bay doors right to the assembly line; supplier personnel unload a few hours of parts, clean up the area and depart. There is no incoming inspection, no staging area, no expediting of material, just a seemingly continuous flow of material. (Hervey, 1982, p 6)

This just-in-time philosophy is generally considered to be one of the major factors underlying the competitive power of the Japanese auto industry over the past three decades. It is natural, therefore, that competitors in other parts of the world have tried to neutralize this competitive edge. Raia (1988) claims that Chrysler's early extensive JIT initiatives were one of the main factors contributing to the recovery in the mid-1980s. The tied-up capital could be reduced by one billion dollars by improving material flow efficiency.

The major changes when adopting JIT deliveries are a decrease in lot sizes and an increase in delivery frequency. In the Honda plant in Ohio there is no tyre inventory at the assembly plant. A local tyre supplier makes 136 deliveries each day by truck. The supplier is responsible for ensuring that the tyre specified is loaded on the conveyor belt connected to the assembly line, according to the schedule decided by the purchaser. This system then works in the same way as the Japanese system discussed above: no storage, no inspection, and no extra handling (Offodile and Arrington, 1992).

Development through supplier relations

Purchasing companies can benefit in a number of ways from cooperation with suppliers in technical development regarding their products and

processes. Two of the most frequently mentioned issues deal with increasing and activating the resource base and shortening lead times in technical development projects.

In close relationships, it is possible to identify and activate complementary resource bases. Many products are characterized by the need for deep knowledge in several areas of technology. Owing to increasing specialization, it is not possible for one company to cover all these areas. It is necessary, therefore, to find suppliers who are willing and able to contribute to technical development. One positive effect will be an extension of the total resource base. Another is that interactive effects might emerge. In the active confrontation of two resource bodies, innovative sparks can arise (Håkansson, 1987).

It is not surprising, therefore, that suppliers have, over time, been increasingly used as a resource in product development. Eriksson and Håkansson (1993) describe two interesting examples of supplier involvement. They deal with development of a robotized system and a piece of medical process equipment. A large number of suppliers are activated in the development processes. In both cases suppliers are responsible for the most innovative parts of the new system.

The other main reason for activating suppliers in development processes is to shorten lead times (see for example Takeuchi and Nonaka, 1986; Burt, 1989; Raia, 1991). Raia presents a number of interesting examples. One is that of Xerox, which has managed to reduce lead times for new products by more than half as a result of increasing supplier involvement. Ingersoll-Rand introduced several suppliers to a project when the new product only was 'a gleam in the eye of the marketing manager'. In this way it is possible to shift certain development activities to suppliers, making it possible to reduce lead times substantially. One important prerequisite for successful implementation of programmes with such aims is that suppliers are involved in the development processes much earlier than they have previously been allowed to be.

Implementation of changes

New forms of supplier relations provide customers with opportunities for rationalization as well as development activities. It is important to observe, however, that such effects do not follow automatically from a concentration to fewer suppliers and policy declarations. On the contrary, a lot of hard work is needed to attain the potential benefits. A number of problems can arise when changes are implemented.

The advantages associated with JIT deliveries can be substantial, but the transition to JIT is far from simple. This is shown by an observation from a purchasing company that could demonstrate that supplies arrived precisely on schedule. On the shop-floor, however, the impression was entirely different.

> But when I walked through the plant I saw weeks of stampings, acres of work-in-process, and subassemblies strewn around the body-shop. Boxes of parts were stacked so high on the chassis and trim lines, that it was difficult to see what was going on in these areas. In fact, the inventory took up so much room that they could have put five major press lines in the same space. This is JIT, I wondered? (Harbour, 1986, p 14)

It is interesting to compare this observation with the attitudes of the American visiting the Japanese company, described earlier. It is very clear that JIT must not be regarded only as a purchasing strategy affecting lot sizes and delivery frequencies. In fact, JIT is a basic determinant of the efficiency of an integrated production system. Activities and requirements directed towards suppliers therefore have few positive effects if they are not followed – or rather preceded – by changes at the buying firms. If this is not done, subdeliveries of the JIT type might increase problems when delivery frequencies are increased.

When JIT was first introduced into the West, the effect seems to have been that inventories were moved to an earlier stage in the production chain. This was found in a comparative study of JIT in Europe, Japan and the USA (Nishiguchi, 1989). In the US automotive industry it was observed that suppliers increased their delivery frequencies. However, these deliveries often came from newly established warehouses. The inventory business in Detroit was booming; one company even picked the name 'JIT Warehousing' (Raia, 1988).

In a major survey of US automotive suppliers, Helper (1991) concluded that suppliers implemented JIT mainly because customers demanded it, but that no major effects were attained in the production processes of suppliers. The prevailing attitude to JIT was that it only transferred responsibility for inventories. Less than one-third of the respondents were of the opinion that JIT decreased the inventory levels for the chain as a whole. The major reason stated for the shortcomings was that customers did not provide suppliers with stable delivery schedules.

The conclusion to be drawn is that efforts to increase efficiency in material flow are major undertakings requiring a number of changes both from suppliers and from the purchasing companies. Therefore it should not be surprising that progress takes time. To establish the links of a total JIT system is a highly demanding process. Toyota began experimenting with its system in 1948. Not until 1965 was a fully synchronized system between Toyota and its first-tier suppliers completed (Nishiguchi, 1987, 1994).

Regarding development cooperation with suppliers, several problems can be identified. One is related to the knowledge of the potential in such collaboration. Burt and Sukoup (1985) concluded that suppliers, at that time, were rather neglected as resources relating to product development. Most customers did not seem to understand the potential available, which the authors considered a major strategic defect. A study of a large number of small and medium-sized firms in Sweden showed that more than one quarter of them had had no cooperative development projects with any supplier during the preceding three-year period (Håkansson, 1989).

What is the explanation for this reluctance towards using supplier resources? One important reason could be a lack of insight into the actual potential. Another reason can be identified as a reluctance on the supplier side. Increasing cooperation should also be of interest to suppliers: through such activities they might be able to strengthen their ties to customers. In spite of this potential advantage, suppliers often appear to hesitate to enter into deeper relations with customers. The reason is that market investments of this type have a cost side that must be compared with the benefits. Suppliers have to invest in specific resources in order to be attractive partners. These resources may be difficult to obtain, as the costs associated with them may be substantial. Furthermore, they sometimes create a very strong dependence, as they might have no alternative use if they are very specialized solutions.

It is also clear that suppliers who are used to customer behaviour such as playing suppliers off against one another will think twice before entering into a close relationship. One prerequisite for increased involvement is some kind of confidence in a long-term business relation. The usual adversarial relationship must be changed to a more symbiotic one. Before this can be achieved, customers themselves have to change their attitudes. Communication of long-term strategic ambitions must be undertaken at the buying company as well as towards the supplier company. It is obvious that changing attitudes is a very time-consuming activity in a large company, especially if the new attitudes deviate substantially from the old ones. Only after these changes within the companies is it possible to change the relationship between them.

Implications

Our analysis of the changing role of purchasing indicates the following tendencies regarding the three strategic issues identified:

(1) an increasing importance of 'buy' as compared with 'make' – i.e. reducing the degree of vertical integration and because the make-or-buy problems are becoming increasingly complex, they deserve more strategic attention;
(2) systematic attempts to structure the suppliers, including a reduction of the supply base, and improvement of coordination among suppliers;
(3) deeper cooperation with individual suppliers to achieve benefits regarding rationalization and technical development.

These changes have naturally brought with them related changes in purchasing and marketing activities. The most dramatic ones have certainly taken place on the purchasing side. The most significant changes have to do with the relationships with individual suppliers, but this has also had important implications for the way purchasing strategies have been formulated and the ways in which purchasing activities have been organized. The marketing function within selling companies has also had to make substantial adaptations but these changes have been easier to integrate into already existing strategies and organizational forms. These changes have also affected the general industry level in terms of production structures and degree of innovativeness. Our final section, therefore, deals with implications for marketing, purchasing and the aggregate industry level.

Purchasing implications

The first and most important implication is that purchasing strategy has become an issue for top management. Never before have so many companies discussed, analysed and formulated offensive strategies for purchasing. The analyses are directed towards finding efficient supplier structures, forming alliances with key suppliers, developing training programmes together with suppliers and activating suppliers in technical development projects. This is a considerable change from the earlier concentration on formulating procedures for efficient purchasing, such as the number of bids that had to be asked for. These changes reflect a new view of purchasing efficiency, which is further analysed in Gadde and Håkansson (1993).

A second implication is related to the organization of purchasing. The major shift has been towards a decentralization of purchasing activities away from the centralized purchasing departments that used to characterize large manufacturing firms. There are two main reasons for this change. The first is that the purchasing decisions must be made by people who are close to the problems to be solved. They must have a good knowledge of the use of the products and about the problems regarding logistical and administrative routines, to be able to capture the potential for rationalization and development. The second reason is that decentralization of purchasing has been well in accord with general organizational trends towards independent profit centres with decentralized responsibility. If purchasing costs account for more than half of the total costs, then each profit centre must be given the right to make its own decisions regarding purchasing and

suppliers. Decentralization makes it generally easier to develop close relationships, while also making it more difficult to coordinate the relationships of the whole company.

A third implication, following the organizational change, is a trend away from functional specialization. The changing role of purchasing will call for a shift towards more integrated problem-solving. Purchasing today is much more focused on technical and logistical matters than before, when strictly commercial issues dominated. Developing supplier relationships means finding efficient solutions, considering the trade-offs between direct costs (mainly price), various indirect costs (in production, administration, material flow, for example) and other strategic benefits from the relationships. This aspect will also affect the role of individual purchasers. Highly specialized purchasers will be replaced by more general problem-solvers. Before, a purchasing officer was often regarded as a rather isolated person responsible only for the commercial aspect of procurement and sometimes considered only as an executor of decisions taken by others. Today, a purchaser must be a member of a team, working closely together with, and coordinating, specialists from other functions. In some companies this change has been so considerable that the purchaser as a functional specialist no longer exists.

Marketing implications

The changes described from the purchasing perspective have, of course, also affected and involved the selling companies. Some suppliers have been very actively involved and have even been change agents, while others have taken part more reluctantly or have even tried to counteract the changes. Those who have considered the transition of purchasing as an opportunity have had to learn a new way of dealing with customers. One characteristic of this new role is the development of mutual relationships, which has two consequences, both contradictory to established marketing traditions. The first is that reaction is as important as action. This means that strategic marketing decisions should increasingly be based on adaptations to the changing purchasing strategies of customers.

The second consequence is that departments other than sales and marketing have to be much more involved in developing the relationships. Direct contacts have to be established between technicians in the two companies, as well as between departments taking care of material and information flows. Furthermore, issues regarding technical development must be discussed between R&D people on both sides. Sometimes even the purchasing department of the selling company needs to be involved in the process, as the customer might have requirements regarding which subsuppliers to use, especially for critical components and materials.

Owing to these conditions, marketing will be much more of an organizational problem than has been previously recognized. The main issue will be to handle the development of individual customer relationships. Developing these relationships will call for coordinating activities from the selling company. The first is that the internal activities of the supplier must be coordinated with those of the customer. The second is that the various customer relationships have to be coordinated in several dimensions. The third, finally, is that the customer relationships have to be coordinated with the selling companies' other external relationships, such as suppliers and technical consultants.

Implications on an aggregate level

The new purchasing philosophy also has effects at a more aggregate level. The development of relationships and networks increases the potential for further specialization. Consequently, companies become increasingly specialized production units, within a network structure. In the future, various production activities will be undertaken by highly specialized companies, which are bound together through close relationships into efficient production structures. As pointed out by Richardson (1993), the combination of a high level of relationship-specific investments and single sourcing might lead to problems with supplier performance, but it is also possible to say that close relationships will increase the pressure on supplier performance. Deepened customer–supplier relationships will be characterized by clear, directed, individually based pressure, as the counterparts are mutually dependent. This also means that the pressure can be expressed in much more detailed and specific terms than is possible in market-mediated pressure. Our conclusion is thus that production structures based on deepened relationships need not be less flexible *per se*.

The transitions in purchasing and buyer–seller relationships also influence the nature of innovation and innovativeness. At an increasing rate, innovations will be developed in the interaction between users and suppliers. In general, this will shorten lead times in product development, which is considered to be an important strategic issue in most industries today.

Another consequence is that innovations will be more in accordance with the needs of the established network structure. It is reasonable to believe that such a change will have positive effects on the failure rate in product innovation. Conversely, a potential negative outcome might be a decrease in the number of revolutionary innovations.

Purchasing's concern for innovativeness is a far cry from its former role. The strategic role of purchasing will emerge naturally as the competitive pressures on businesses increase.

References

Axelsson, B. and Håkansson, H. (1984) Inköp för konkurrenskraft, Stockholm: Liber

Blois, K. (1972) 'Vertical quasi-integration', *Journal of Industrial Economics*, 20 (3), pp 33–41

Burt, D. (1989) 'Managing suppliers up to speed', *Harvard Business Review*, July–August, pp 127–135

Burt, D. and Sukoup, W. (1989) 'Purchasing's role in new product development', *Harvard Business Review*, September–October, pp 90–97

Carlisle J. and Parker, R. (1989) *Beyond Negotiation, Redeeming Customer–Supplier Relationships*, Chichester: John Wiley and Sons

Culliton, J. (1942) *Make-or-buy*, Boston: Harvard University Graduate School of Business Administration

Dillforce, W. (1986) 'Purchasing – a singular way to increase competitiveness', *Financial Times*, 24 October

Dirrheimer, M. and Hübner, T. (1983) 'Vertical integration and performance in the automotive industry', paper presented at the Future of Automobile Forum, Boston: MIT

Dubois, A., Gadde, L.-E. and Håkansson, H. (1989) 'The impact of information technology on purchasing behaviour and supplier markets', Working Paper No. 1989:11, Gothenburg: Institute for Management of Innovation and Technology

Dubois, A. and Håkansson, H. (1993) 'Relationships as activity links', *Proceedings of the conference Forms of Interorganizational Networks: Structures and Processes*, Berlin: European Science Foundation (in press)

Eriksson, A.-K. and Håkansson, H. (1993) 'Getting innovations out of supplier networks', *Journal of Business-to-Business Marketing*, 1(3), pp 3–34

Ford, D., Cotton, B., Farmer, D., Gross, A. and Wilkinson, I. (1993) 'Make-or-buy decisions and their implications', *Industrial Marketing Management*, 22, pp 207–214

Frazier, G., Spekman, R. and O'Neal, C. (1988) 'Just-in-time exchange relationships in industrial markets', *Journal of Marketing*, 52 (October), pp 52–67

Gadde, L.-E. and Grant, B. (1984) 'Quasi-integration, supplier networks and technological cooperation in the automotive industry', *Proceedings of the International Research Seminar on Industrial Marketing*, Stockholm: Stockholm School of Economics

Gadde, L.-E. and Håkansson, H. (1993) *Professional Purchasing*, London: Routledge

Håkansson, H. (ed) (1987) *Industrial Technological Development. A Network Approach*, Beckenham: Croom Helm

Håkansson, H. (1989) *Corporate Technological Behaviour – Cooperation and Networks*, London: Routledge

Harbour, J. (1986) 'What is just-in-time manufacturing', *Automotive Industries*, January, p 14

Hayes, R. and Abernathy, W. (1980) 'Managing our way to economic decline', *Harvard Business Review*, July–August, pp 67–77

Helper, S. (1991) 'How much has really changed between US automakers and their suppliers', *Sloan Management Review*, Summer, pp 15–28

Hervey, R. (1982) 'Preliminary observation on manufacturer–supplier relations in the Japanese automotive industry', The Joint US–Japan Automotive Study, Working Paper series No. 5, Ann Arbor: University of Michigan

Janch, L. and Wilson, H. (1979) 'A strategic perspective for make or buy decisions', *Long Range Planning*, 12 (December), pp 56–61

Kumpe, T. and Bolwijn, P. (1988) 'Manufacturing: the new case for vertical integration', *Harvard Business Review*, March–April, pp 75–81

Lamming, R. (1993) *Beyond Partnership. Strategies for Innovation and Lean Supply*, Hemel Hempstead, UK: Prentice-Hall

Leenders, M. and Nollet, J. (1984) 'The grey zone in make or buy', *Journal of Purchasing and Materials Management*, Fall, pp 10–15

Marler, D. (1989) 'The post Japanese model of automotive component supply: selective North American case studies', IMVP International Policy Forum, Boston: MIT

Miles, R. and Snow, C. (1986) 'Organizations: new concepts for new forms', *California Management Review*, 18 (3), pp 62–73

Morgan, I. (1987) 'The purchasing revolution', *The McKinsey Quarterly*, Spring, pp 49–55

Newman, R. (1988) 'Single source qualification', *Journal of Purchasing and Materials Management*, Summer, pp. 10–17

Nishiguchi, T. (1984) 'Competing systems of automotive components supply: an examination of the Japanese "clustered control" model and the "Alps" structure', First Policy Forum, International Motor Vehicle Program, Boston: MIT

Nishiguchi, T. (1989) 'Is JIT really JIT?', Third Policy Forum, International Motor Vehicle Program, Boston: MIT

Nishiguchi, T. (1993) *Strategic Industrial Sourcing: the Japanese Advantage*, New York: Oxford University Press

Offodile, F. and Arrington, D. (1992) 'Support of successful just-in-time implementation: the changing role of purchasing', *International Journal of Physical Distribution and Logistics Management*, 22 (5), pp 38–46

Raia, E. (1988) 'JIT in Detroit', *Purchasing*, 15 September, pp 68–77

Raia, E. (1991) 'Taking time out of product design', *Purchasing*, 4 (April 4), pp 36–39

Raia, E. (1993) 'The extended enterprise', *Purchasing*, 4 March, pp 48–51

Richardson, J. (1993) 'Parallel sourcing and supplier performance in the Japanese automobile industry', *Strategic Management Journal*, 14, pp 339–350

Spekman, R. (1985) 'Strategic supplier selection: understanding long-term buyer relationships', *Business Horizons*, July–August, pp 75–81

Takeuchi, H. and Nonaka, I. (1986) 'The new new-product development game', *Harvard Business Review*, January–February, pp 137–146

Venkatesan, R. (1992) 'Strategic sourcing: to make or not to make', *Harvard Business Review*, November–December, pp 98–108

Womack, J., Jones, D. and Roos, D. (1990) *The Machine that Changed the World*, New York: Rawson Associates